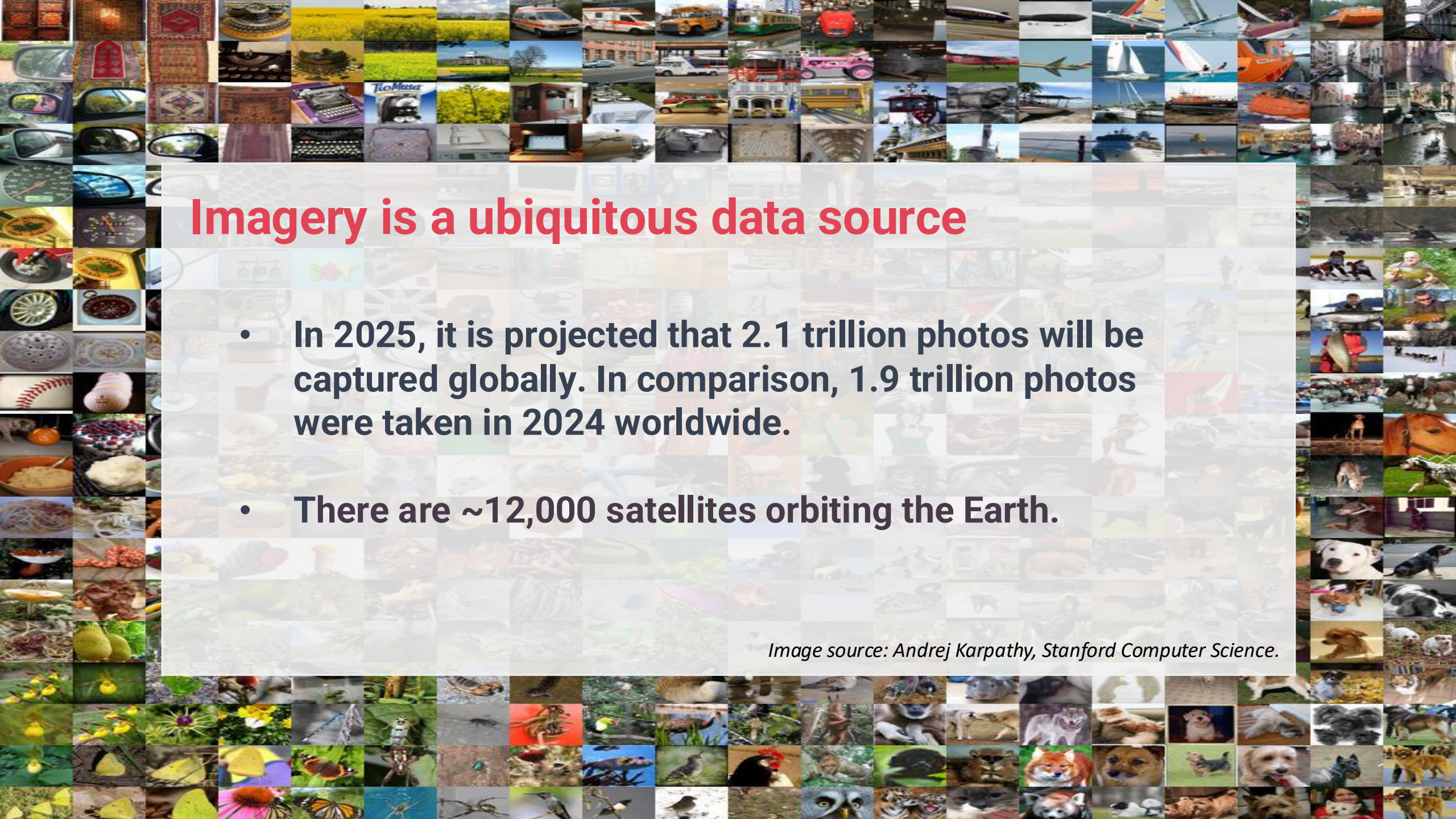
# Geospatial Artificial Intelligence in Evaluation Practice

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IEG Independent Evaluation Group | WORLD BANK GROUP World Bank • IFC • MIGA

# 1 | Motivation



# Imagery is a ubiquitous data source

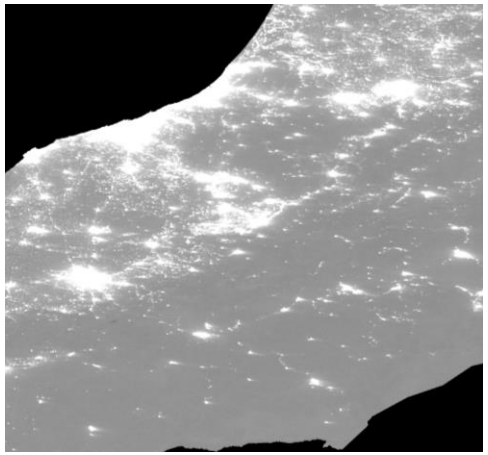
- In 2025, it is projected that 2.1 trillion photos will be captured globally. In comparison, 1.9 trillion photos were taken in 2024 worldwide.
- There are ~12,000 satellites orbiting the Earth.

*Image source: Andrej Karpathy, Stanford Computer Science.*

# There is a wealth of imagery available for Geospatial Analysis



Optical Daylight Satellite Image
(Source: Google Earth)



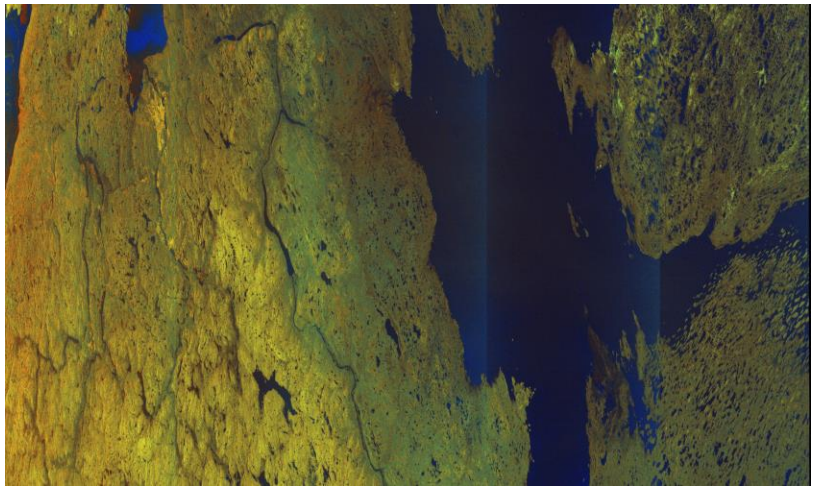
Optical Nighttime Satellite Image
(Source: VIIRS)



Streetscape Digital Photo
(Source: Mapillary)

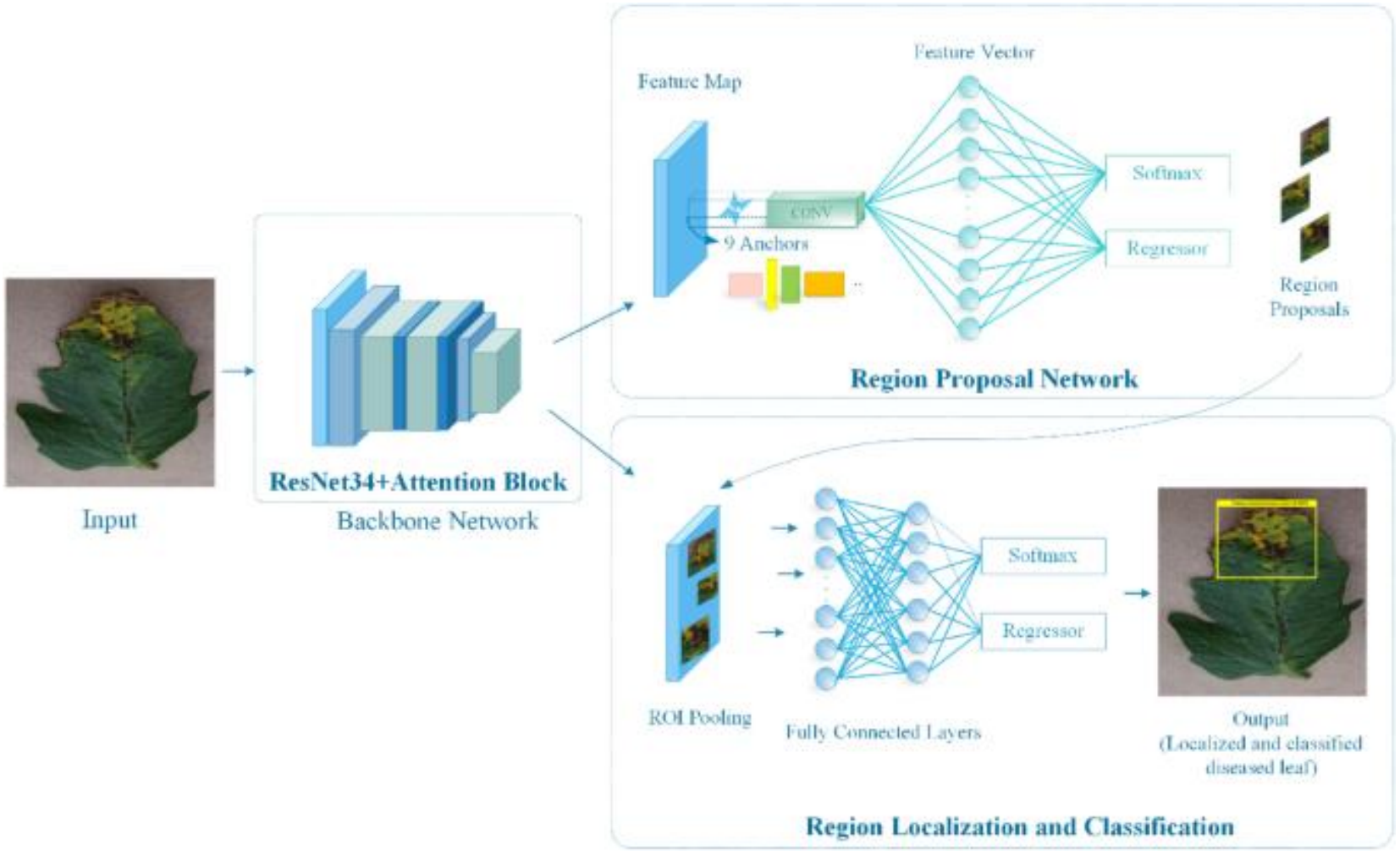


Aerial (Drone) Imagery
(Source: Dar Rapid Transit Agency)



Synthetic Aperture Radar (SAR) Image
(Source: ERS-2, ASF DAAC)

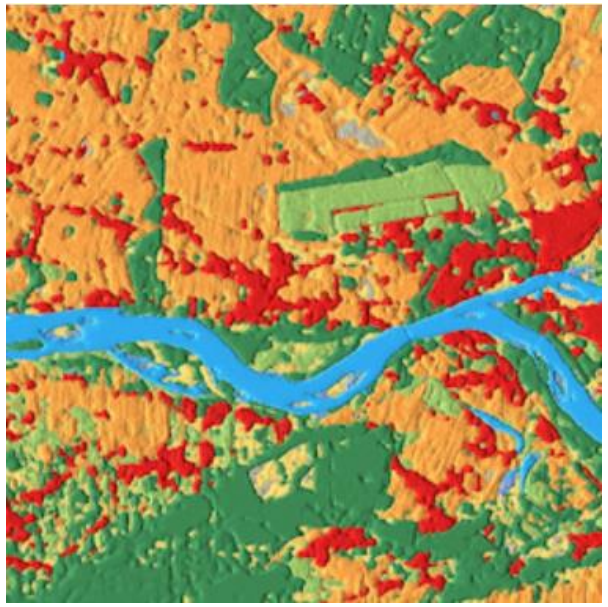
# GeoAI = Artificial Intelligence + Geospatial Data




*Reference: Nawaz, Marriam, Tahira Nazir, Ali Javed, Momina Masood, Junaid Rashid, Jungeun Kim, and Amir Hussain. "A robust deep learning approach for tomato plant leaf disease localization and classification." Scientific reports 12, no. 1 (2022): 18568.*



Microsoft Bing Buildings Footprint Dataset



Dynamic World Near Real Time Landcover Dataset

# AI-Generated Datasets

- There is a growing number of geospatial datasets generated using AI—ranging from global land cover maps to building footprints and road networks.
- These AI-generated datasets offer ready-to-use, high-resolution spatial information that can support a wide range of applications, from urban planning to disaster response.



Semantic segmentation using convolutional neural networks.
Source: UPSNet-101-M Cityscapes.

# AI Modelling

- Pre-trained and ready to use models
- Transfer learning and model fine-tuning
- Fully custom models

# 2 | Applications in Evaluations

## Example 1: AI-generated datasets

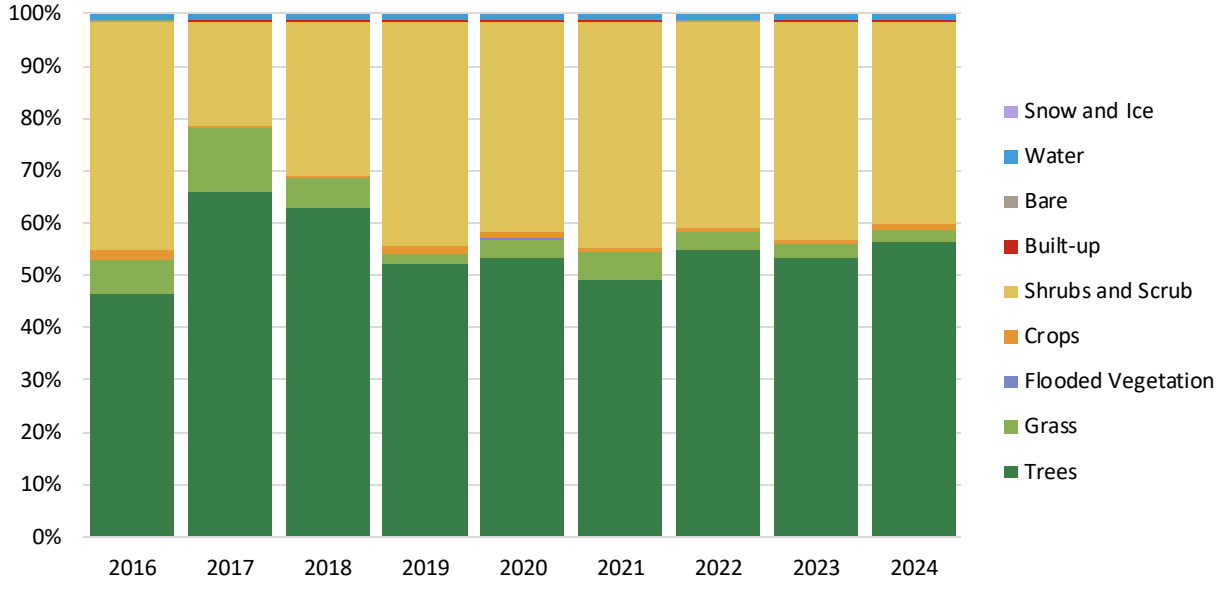
# Biodiversity Evaluation

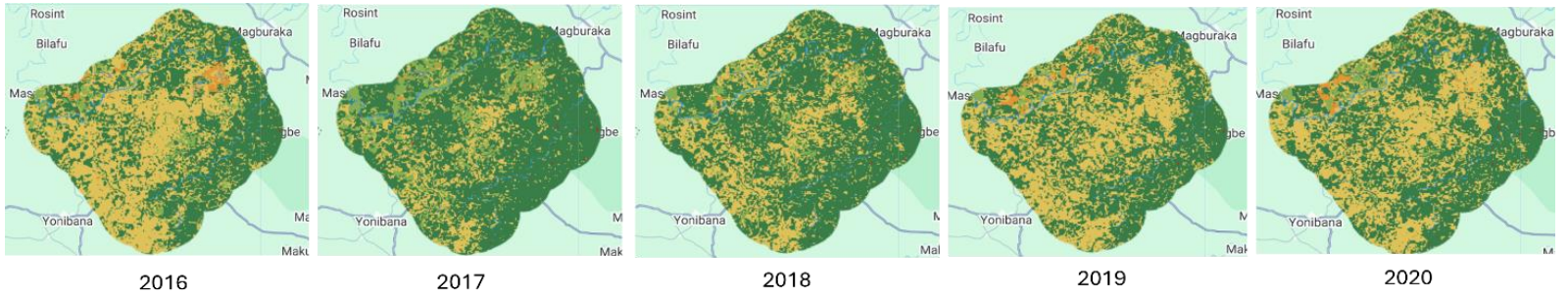
**Evaluation question:**
How well are biodiversity projects achieving their biodiversity goals?

**Unit of analysis:**
526 terrestrial or hybrid protected areas.

**Dataset used:** Dynamic World.

*Annual Land Cover Composition in Mamunta-Mayosso, Sierra Leone (2016–2024)*



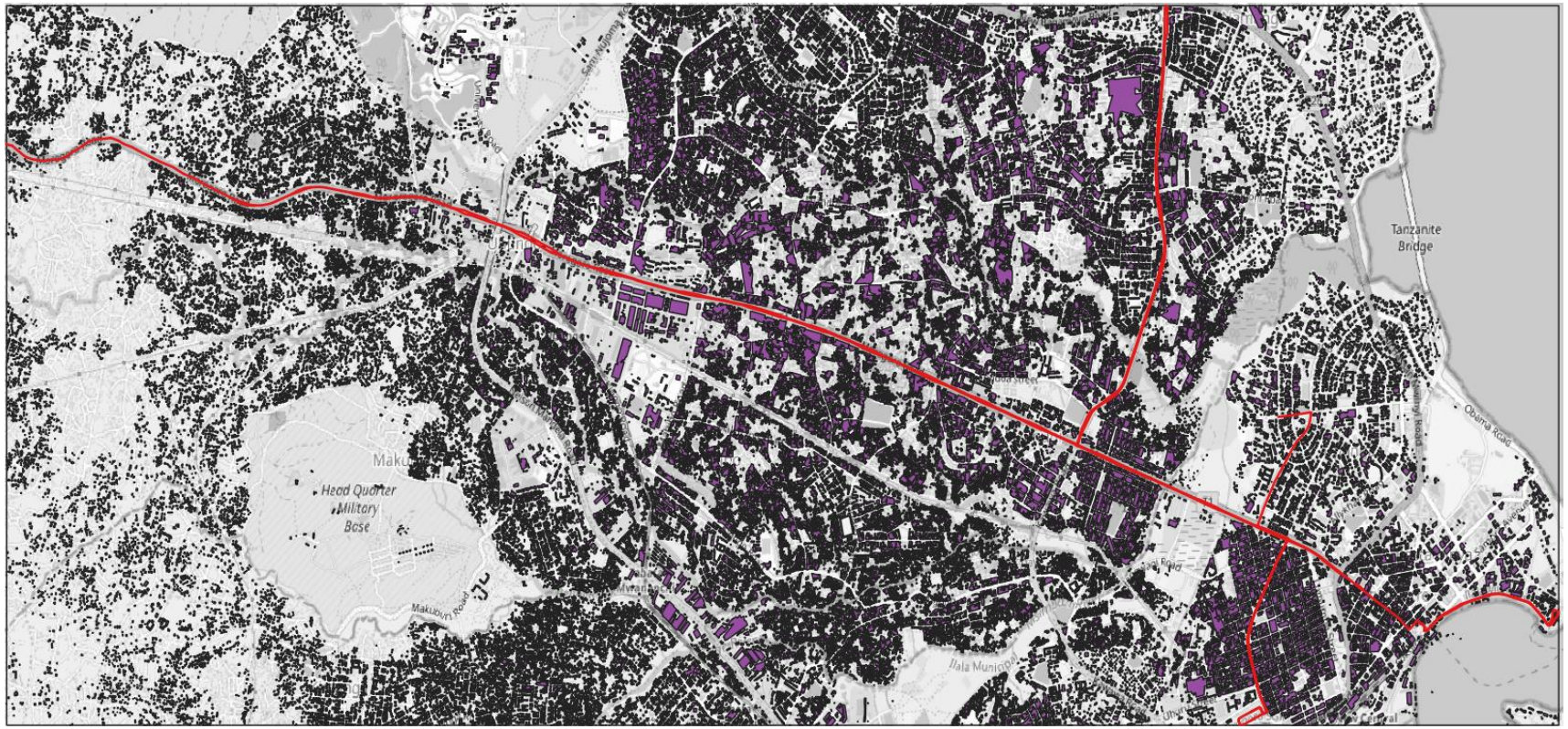
## Example 2: AI-generated datasets

# Tanzania Country Program Evaluation (1)

**Evaluation question:**
To what extent has the World Bank Group contributed to spatial transformation by improving intraurban as well as urban-rural linkages and facilitating regional connectivity.

**Unit of analysis:**
Dar Es Salaam Bus Rapid Transit Phase 1 Area.

**Dataset used:** Microsoft/Bing Building Footprints.



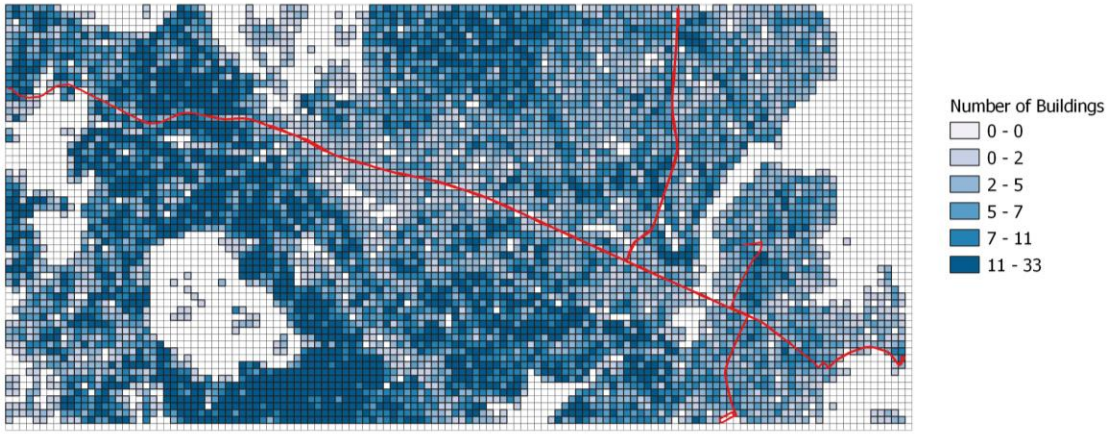

The map above represents individual building parcels overlaid to an OpenStreetMap basemap. The BRT route (Phase I) is also displayed in the map in red.
The area comprises 48,050 parcels covering 14.55 $km^2$
The size of individual parcels range between 0.2 $m^2$ and 81,746 $m^2$ (mean = 303 $m^2$).

## Example 2: AI-generated datasets

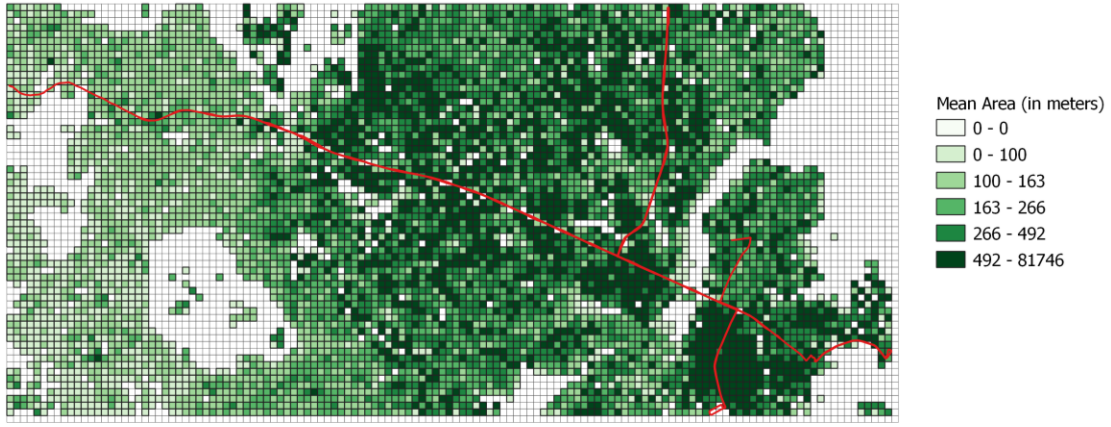
# Tanzania Country Program Evaluation (2)

- 5 different metrics which quantify aspects of the parcels' morphology were defined.
- A 100x100 meter grid was created and overlaid to the parcels' data. Each of the morphology metrics were calculated and assigned to grid cell where the centroid of each parcel is located.
- Grid cells were clustered based on the values of each of the 5 morphology metrics using Gaussian Mixture Modelling approach.

### Metric 1: Building Density



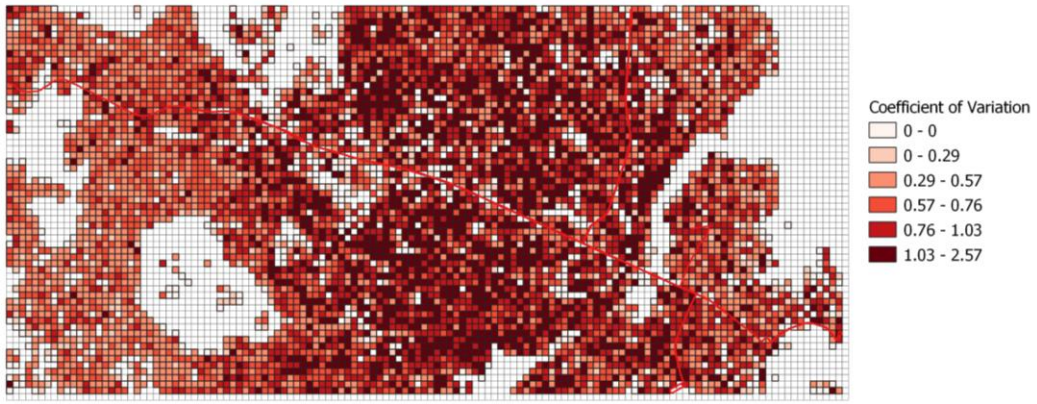

### Metric 2: Mean Building Area



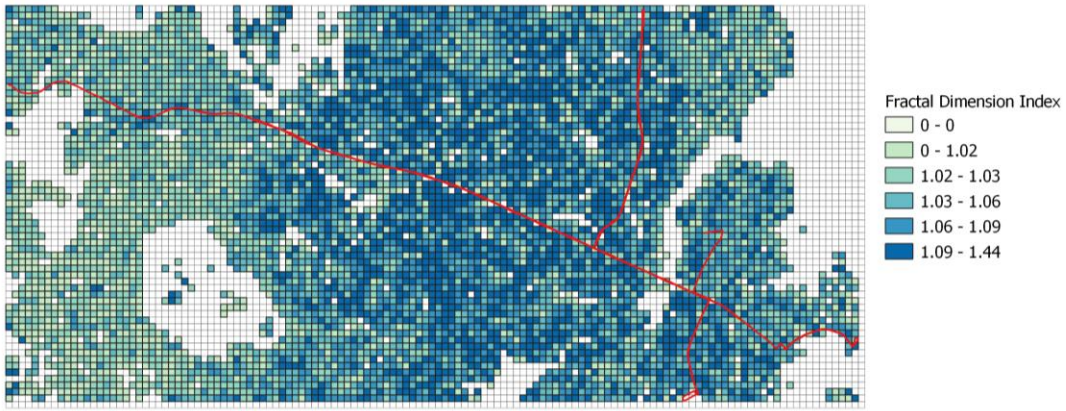
## Example 2: AI-generated datasets

# Tanzania Country Program Evaluation (3)

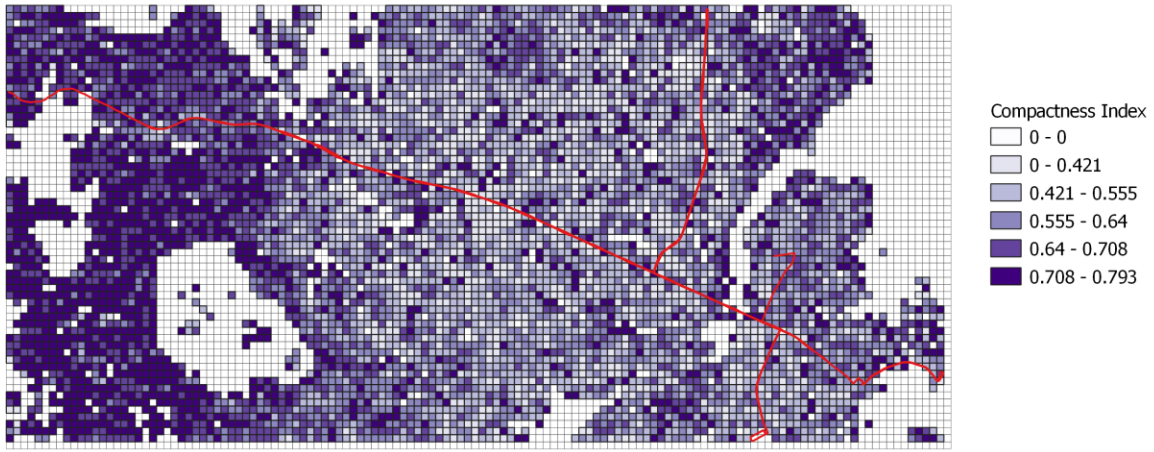
### Metric 3: Building Size Coefficient of Variation




### Metric 4: Fractal Dimension Index




### Metric 5: Compactness Index

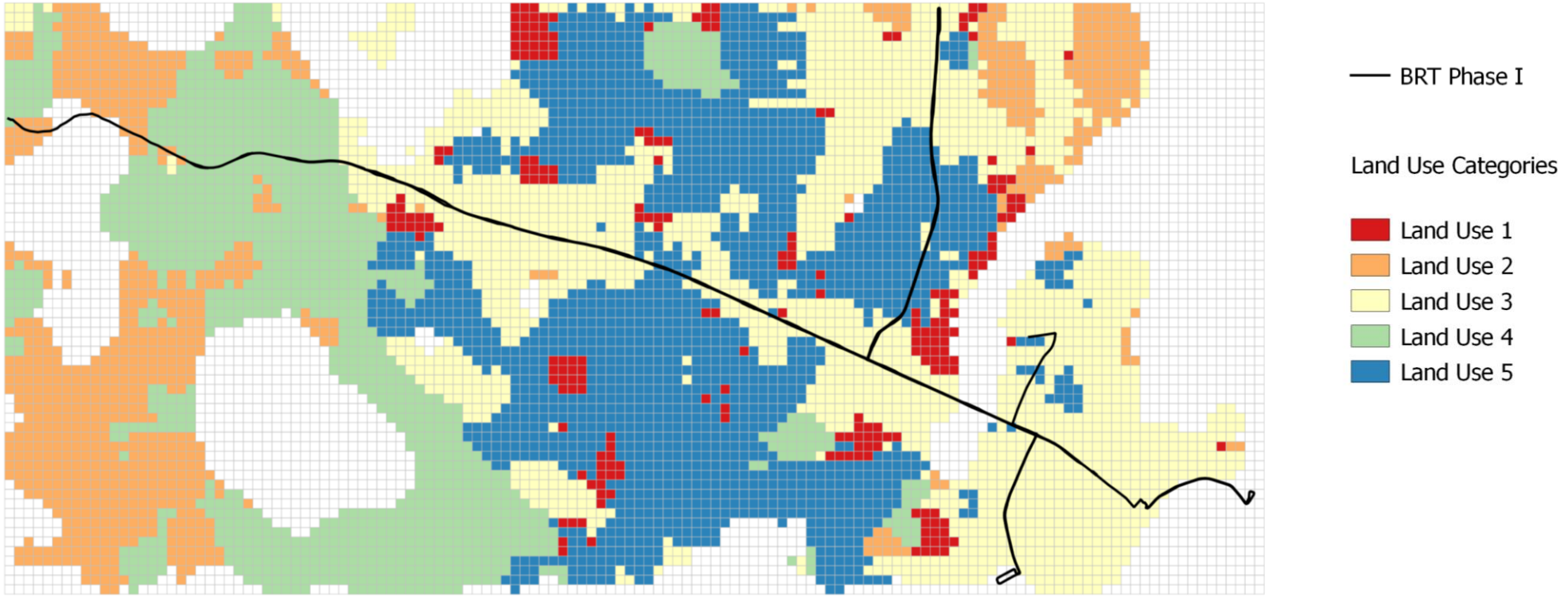


## Example 2: AI-generated datasets

# Tanzania Country Program Evaluation (4)

### Estimated Land Use Categories based on Parcels' Morphology

Land use classes were estimated using a Gaussian Mixture Model (GMM) with 5 clusters. The number of clusters was determined using the Akaike Information Criterion (AIC). Predictions were smoothed using a majority filter with a radius of 2.

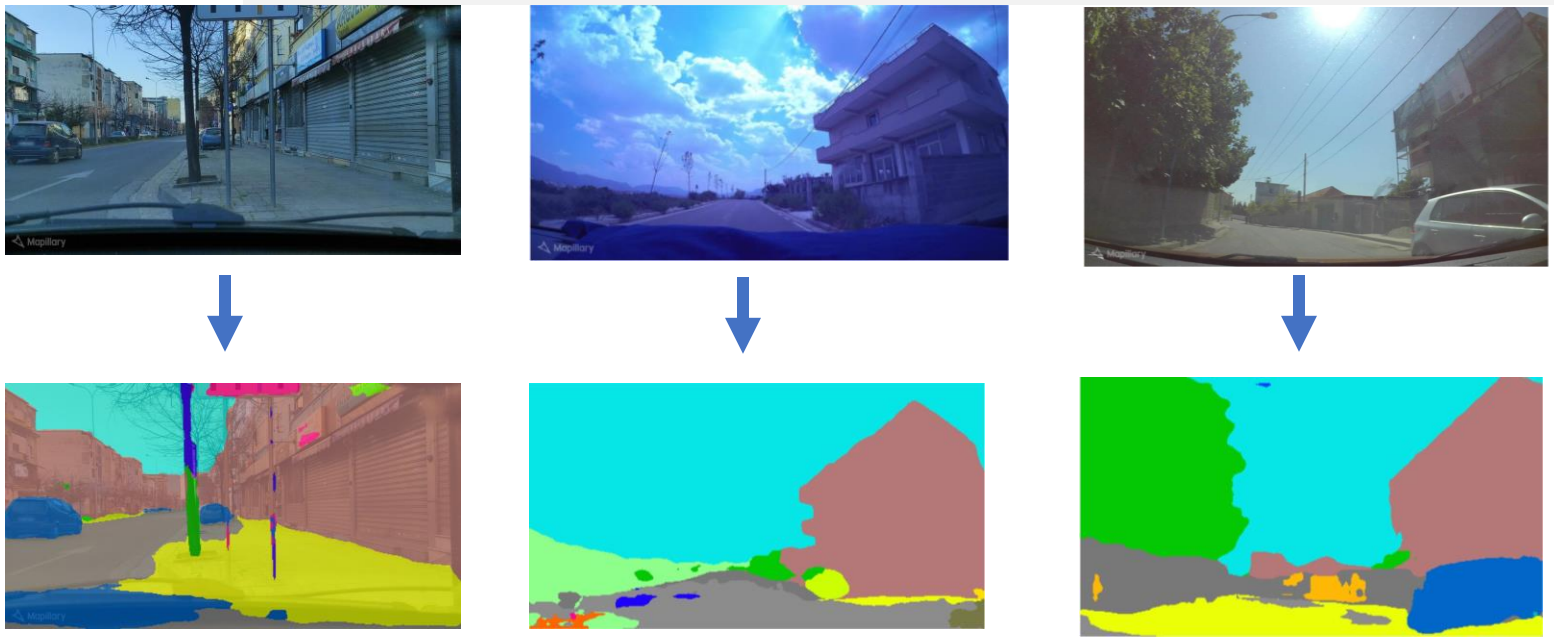


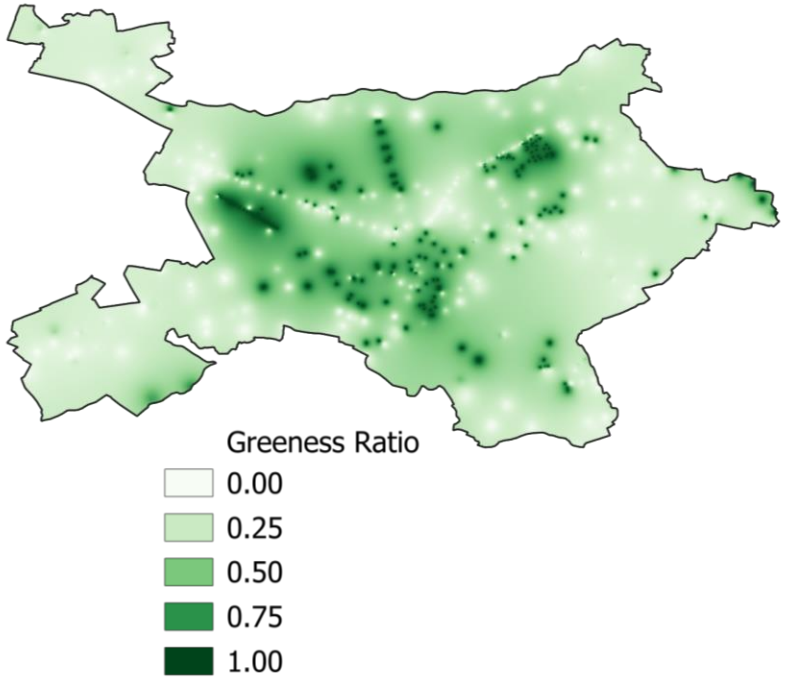
## Example 3: AI modelling

# Tirana Learning Engagement

**Evaluation question:**
Has the urban environment improved in targeted neighborhoods following infrastructure investments? (urban livability)

**Unit of analysis:**
Supported areas in Tirana (Albania).






*Location: Tirana (Albania). Data source: Mapillary + additional photos taken by the team.*

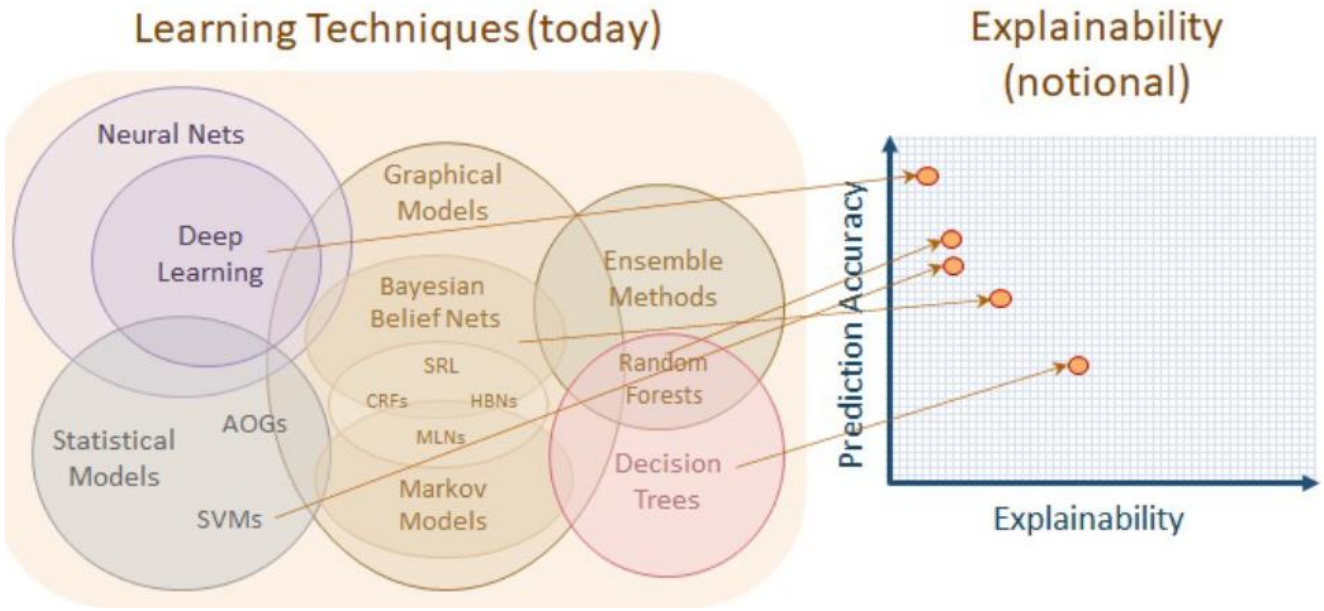
# 3 | Ethical Considerations



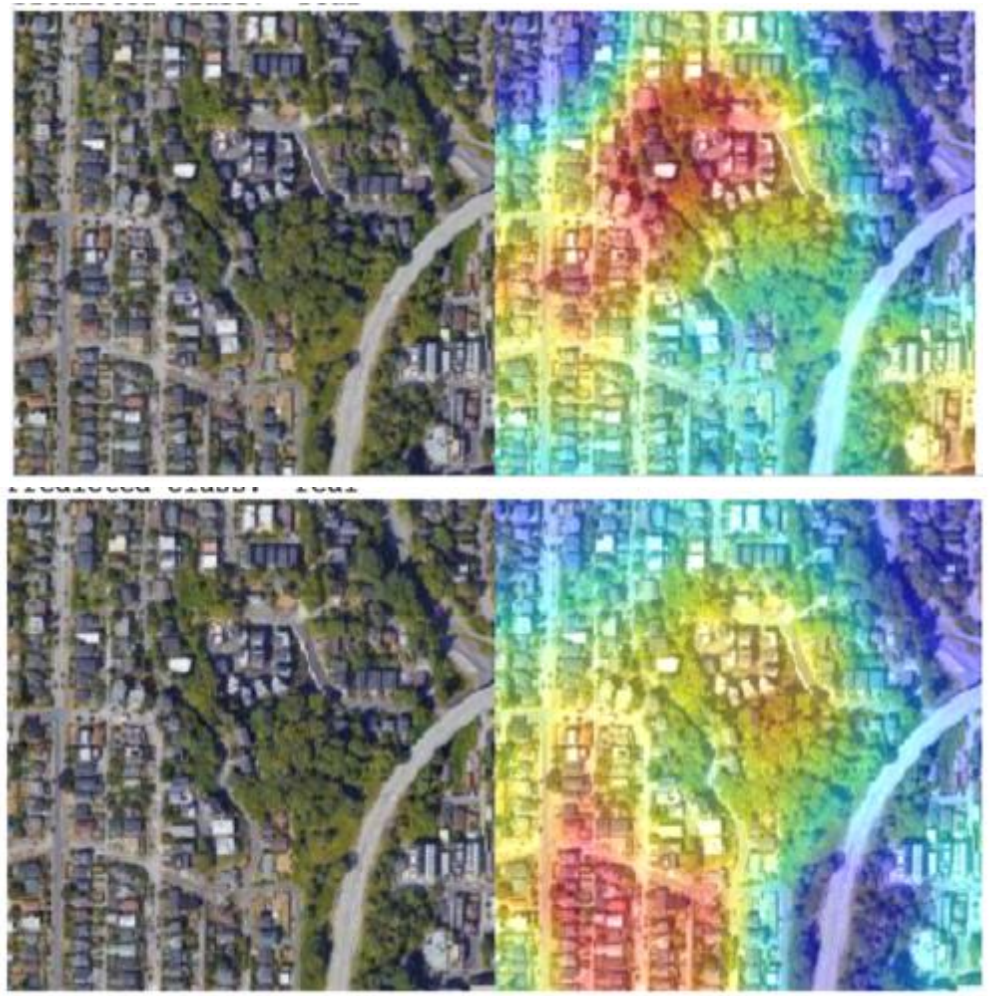
# Ethical Considerations

- GeoAI models often operate as black boxes—users see predictions, not the reasoning behind them.
- Biases in training data and geographic mismatch can lead to misleading or unjust outcomes.
- Proxies can mislead without validations (e.g. NTL).
- Pipelines can hide system errors.

# Opening the Black Box




(Gunning et al.)



GRAD-CAM Activations



# Mitigations Strategies

- Document models: use model cards and data sheets.
- Audit for bias: especially in underrepresented areas.
- Communicate uncertainty: avoid false precision.
- Validate locally: engage experts and communities.

# Thank You!